

Student Utility Program

Java Lab Record

1. Aim

To develop a menu-driven Student Utility Program in Java that demonstrates basic Java concepts such as methods, parameters, return values, conditional statements, switch-case, loops, arrays, and user input using Scanner.

2. Step-by-Step Procedure

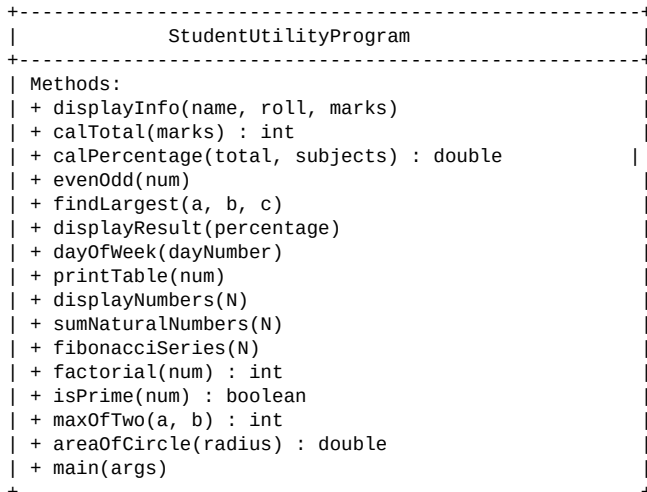
1. Create a Java class named StudentUtilityProgram.
2. Import the Scanner class to accept input from the user.
3. Create methods for different student utility operations.
4. Create the displayInfo() method to display student details, total marks, percentage, and grade.
5. Create methods for calculating total marks and percentage.
6. Create methods for even/odd checking, finding the largest of three numbers, displaying grades, and finding the day of the week.
7. Create loop-based methods for multiplication table, displaying numbers from 1 to N, sum of natural numbers, and Fibonacci series.
8. Create methods with parameters and return values for factorial, prime checking, maximum of two numbers, and area of a circle.
9. In the main() method, display a menu containing all available operations.
10. Read the user's choice using Scanner.
11. Use switch-case to call the corresponding method.
12. Repeat the menu using a do-while loop until the user selects 0.
13. Close the Scanner object after the program terminates.

3. Algorithm

1. Start.
2. Create a Scanner object.
3. Display the Student Utility Program menu.
4. Read the user's choice.
5. If choice is 1, read student details and marks, then calculate total, percentage, and grade.
6. If choice is 2, check whether the number is even or odd.
7. If choice is 3, find the largest among three numbers.
8. If choice is 4, display the grade based on percentage.
9. If choice is 5, display the day corresponding to the day number.
10. If choice is 6, display the multiplication table.

11. If choice is 7, display numbers from 1 to N.
12. If choice is 8, calculate the sum of first N natural numbers.
13. If choice is 9, display the Fibonacci series.
14. If choice is 10, calculate the factorial.
15. If choice is 11, check whether the number is prime.
16. If choice is 12, find the maximum of two numbers.
17. If choice is 13, calculate the area of a circle.
18. If choice is 0, exit the program; otherwise display invalid choice.
19. Repeat the menu until choice is 0.
20. Close Scanner.
21. Stop.

4. UML Class Diagram



The class contains methods for student information, conditional operations, loop-based operations, and methods that return calculated values.

5. Code

```
package studentutility;
import java.util.Scanner;

public class StudentUtilityProgram {
    // Part A: basics
    public static void displayInfo(String name, int roll, int[] marks) {
        int total = calTotal(marks);
        double percentage = calPercentage(total, marks.length);
        System.out.println("\nStudent information : ");
        System.out.println("Name of the student: " + name);
        System.out.println("Roll number: " + roll);
        System.out.println("Total marks: " + total);
        System.out.println("Percentage: " + percentage);
        displayResult(percentage);
    }

    public static int calTotal(int[] marks) {
        int total = 0;
        for (int m : marks)
            total += m;
        return total;
    }

    public static double calPercentage(int total, int subjects) {
        return (double) total / subjects;
    }

    // Conditional statements
    public static void evenOdd(int num) {
        System.out.println(num % 2 == 0 ? num + " is Even" : num + " is Odd");
    }

    public static void findLargest(int a, int b, int c) {
        int largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);
        System.out.println("Largest number is: " + largest);
    }

    public static void displayResult(double percentage) {
        if (percentage >= 90) System.out.println("Grade: A+");
        else if (percentage >= 75) System.out.println("Grade: A");
        else if (percentage >= 60) System.out.println("Grade: B");
        else if (percentage >= 40) System.out.println("Grade: C");
        else System.out.println("Grade: Fail");
    }

    public static void dayOfWeek(int dayNumber) {
        switch (dayNumber) {
            case 1: System.out.println("Monday"); break;
            case 2: System.out.println("Tuesday"); break;
        }
    }
}
```

```

        case 3: System.out.println("Wednesday"); break;
        case 4: System.out.println("Thursday"); break;
        case 5: System.out.println("Friday"); break;
        case 6: System.out.println("Saturday"); break;
        case 7: System.out.println("Sunday"); break;
        default: System.out.println("Invalid day number!");
    }
}

// Loops
public static void printTable(int num) {
    System.out.println("Multiplication Table of " + num);
    for (int i = 1; i <= 10; i++) {
        System.out.println(num + " x " + i + " = " + (num * i));
    }
}

public static void displayNumbers(int N) {
    for (int i = 1; i <= N; i++) System.out.print(i + " ");
    System.out.println();
}

public static void sumNaturalNumbers(int N) {
    int sum = 0;
    for (int i = 1; i <= N; i++) sum += i;
    System.out.println("Sum of first " + N + " natural numbers = " + sum);
}

public static void fibonacciSeries(int N) {
    int a = 0, b = 1;
    System.out.print("Fibonacci Series: ");
    for (int i = 1; i <= N; i++) {
        System.out.print(a + " ");
        int next = a + b;
        a = b;
        b = next;
    }
    System.out.println();
}

// Methods: with parameters and return value
public static int factorial(int num) {
    int fact = 1;
    for (int i = 1; i <= num; i++) fact *= i;
    return fact;
}

public static boolean isPrime(int num) {
    if (num <= 1) return false;
    for (int i = 2; i <= Math.sqrt(num); i++) {
        if (num % i == 0) return false;
    }
    return true;
}

public static int maxOfTwo(int a, int b) {
    return (a > b) ? a : b;
}

public static double areaOfCircle(double radius) {
    return Math.PI * radius * radius;
}

// Call each method from main
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int choice;

    do {
        System.out.println("\n--- Student Utility Program ---");
        System.out.println("1. Student Info");
        System.out.println("2. Even/Odd Check");
        System.out.println("3. Largest of Three Numbers");
        System.out.println("4. Display Grade");
        System.out.println("5. Day of Week");
        System.out.println("6. Multiplication Table");
        System.out.println("7. Display Numbers 1 to N");
        System.out.println("8. Sum of Natural Numbers");
        System.out.println("9. Fibonacci Series");
        System.out.println("10. Factorial");
    }
}

```

```

System.out.println("11. Prime Check");
System.out.println("12. Max of Two Numbers");
System.out.println("13. Area of Circle");
System.out.println("0. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();

switch (choice) {
    case 1:
        sc.nextLine();
        System.out.print("Enter Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Roll Number: ");
        int roll = sc.nextInt();
        int[] marks = new int[3];
        System.out.println("Enter 3 subject marks:");
        for (int i = 0; i < 3; i++) marks[i] = sc.nextInt();
        displayInfo(name, roll, marks);
        break;

    case 2:
        System.out.print("Enter number: ");
        evenOdd(sc.nextInt());
        break;

    case 3:
        System.out.print("Enter three numbers: ");
        findLargest(sc.nextInt(), sc.nextInt(), sc.nextInt());
        break;

    case 4:
        System.out.print("Enter percentage: ");
        displayResult(sc.nextDouble());
        break;

    case 5:
        System.out.print("Enter day number (1-7): ");
        dayOfWeek(sc.nextInt());
        break;

    case 6:
        System.out.print("Enter number: ");
        printTable(sc.nextInt());
        break;

    case 7:
        System.out.print("Enter N: ");
        displayNumbers(sc.nextInt());
        break;

    case 8:
        System.out.print("Enter N: ");
        sumNaturalNumbers(sc.nextInt());
        break;

    case 9:
        System.out.print("Enter N: ");
        fibonacciSeries(sc.nextInt());
        break;

    case 10:
        System.out.print("Enter number: ");
        int num = sc.nextInt();
        System.out.println("Factorial = " + factorial(num));
        break;

    case 11:
        System.out.print("Enter number: ");
        int primeNum = sc.nextInt();
        System.out.println(primeNum +
            (isPrime(primeNum) ? " is Prime" : " is Not Prime"));
        break;

    case 12:
        System.out.print("Enter two numbers: ");
        System.out.println("Max = " +
            maxOfTwo(sc.nextInt(), sc.nextInt()));
        break;

    case 13:

```

```
        System.out.print("Enter radius: ");
        System.out.println("Area = " +
            areaOfCircle(sc.nextDouble()));
        break;

    case 0:
        System.out.println("Exiting program...");
        break;

    default:
        System.out.println("Invalid choice!");
    }
} while (choice != 0);

sc.close();
}
```